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MZ 505
Inventory Systems

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Session 02: EOQ Model

MIT GLOBAL SCALE NETWORK

Inventory decisions

- There are two fundamental **decisions** in inventory management:
  - How much to order?
  - When to order?
- The main **objective** is to minimize total inventory costs.
- Some key **parameters** to consider:
  - Cost of the items (purchase or material cost)
  - Cost of placing an order
  - Cost of carrying inventory
  - Cost of shortage /stockout
  - Lead time
  - Demand variability

The Atlantic Coast Tire Corporation problem

The Atlantic Coast Tire Corporation (ACT) is the east coast distributor of Eversafe tires, supplying 1500 retail stores and service stations. Currently, ACT follows a simple policy for replenishing its inventory. When the inventory level of a particular size tire gets low, ACT places a large order with Eversafe to replenish its stock. Shipments arrive by truck 9 working days after the placement of an order.

The following are data for tire R13:

- These tires sell at a regular rate of about 500 per month.
- The current policy has been to order 1,000 tires as needed every couple months.
- The order is placed just in time to have the delivery arrive as inventory runs out.

The Atlantic Coast Tire Corporation problem

Inventory **cost components** for R13:

- The purchase price from Eversafe is $20 per tire.
- The cost for placing an order is $115.
- The annual cost of holding inventory is $4.20 per tire.
- The cost of being out of stock is $7.50 per tire short.

Questions:

- What is the total annual cost of the current policy for ACT?
- What is the annual variable cost of the current policy for ACT?
- Is the policy followed optimal? If not, how we could derive the optimal policy?

Cost components of inventory models

- Purchasing Cost (C). The direct cost of replenishing inventory, whether through purchasing or manufacturing of the product.
  - For the ACT Example: C = $20 per tire
- Setup/Ordering Cost (S). The cost of setting up the manufacturing process for another production run (for a manufacturer) or the various costs associated with transportation, receiving, and processing of an order (for a buyer)
  - For the ACT Example: S = $115 per order
- Holding Cost (H). The cost of holding units in inventory (obsolescence, handling, insurance, occupancy, financial)
  - For the ACT Example, H = $4.20 per unit per year
- Shortage Cost (p). The cost incurred when there is a need for the product, but none available in inventory, resulting in e.g. lost current and future sales, expediting delivery, etc.
  - For the ACT Example, p = $7.50 per unit short per year

EOQ model

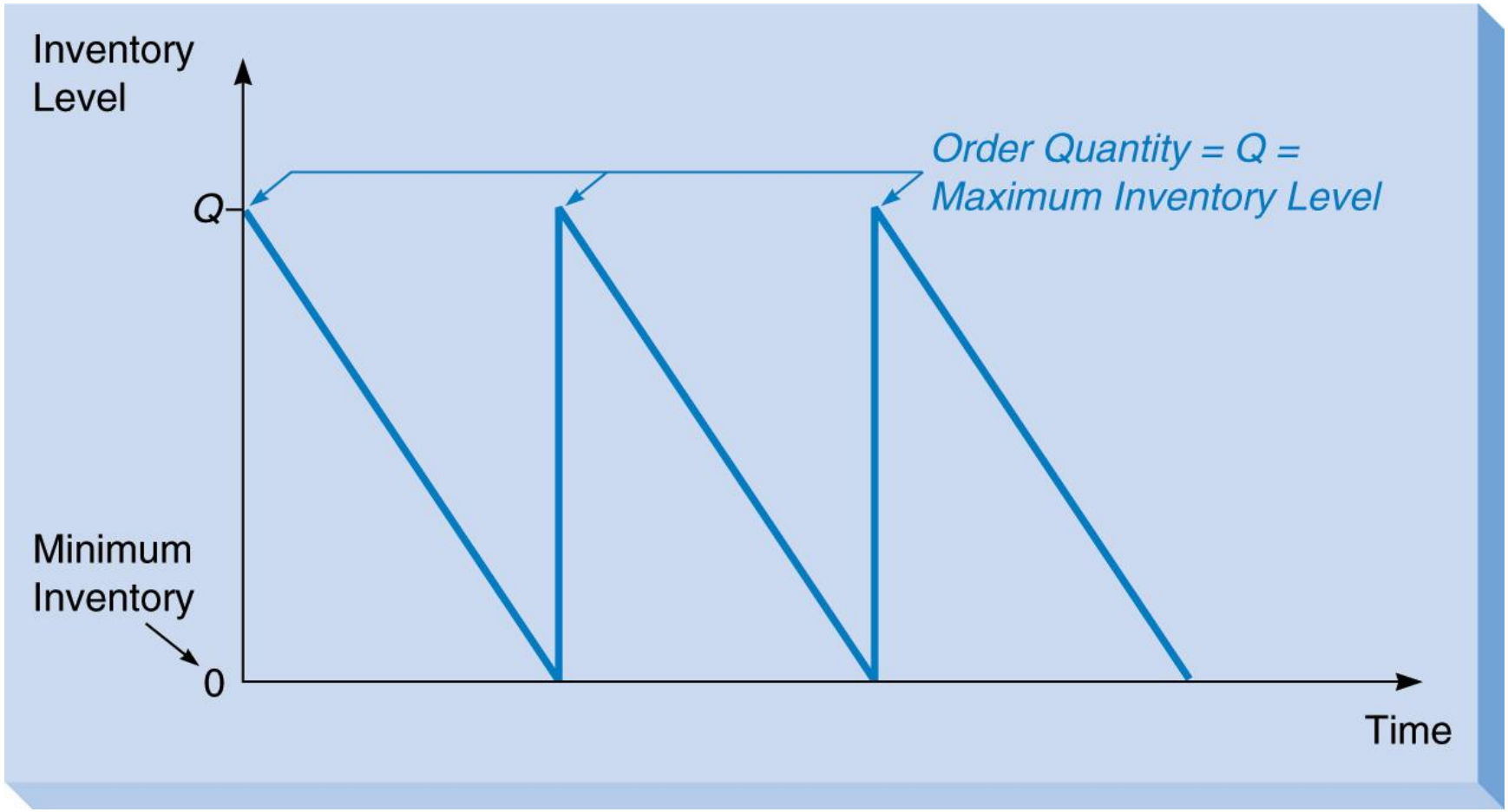
Economic Order Quantity (EOQ)

- Earliest quantitative inventory model
  - Harris 1913, Wilson 1934
- Batch size: A decision regarding **economies of scale**
- Fundamental tradeoff
  - Ordering cost versus inventory carrying cost
- Both practical and theoretical value
- Based on a number of assumptions

Classical EOQ model assumptions

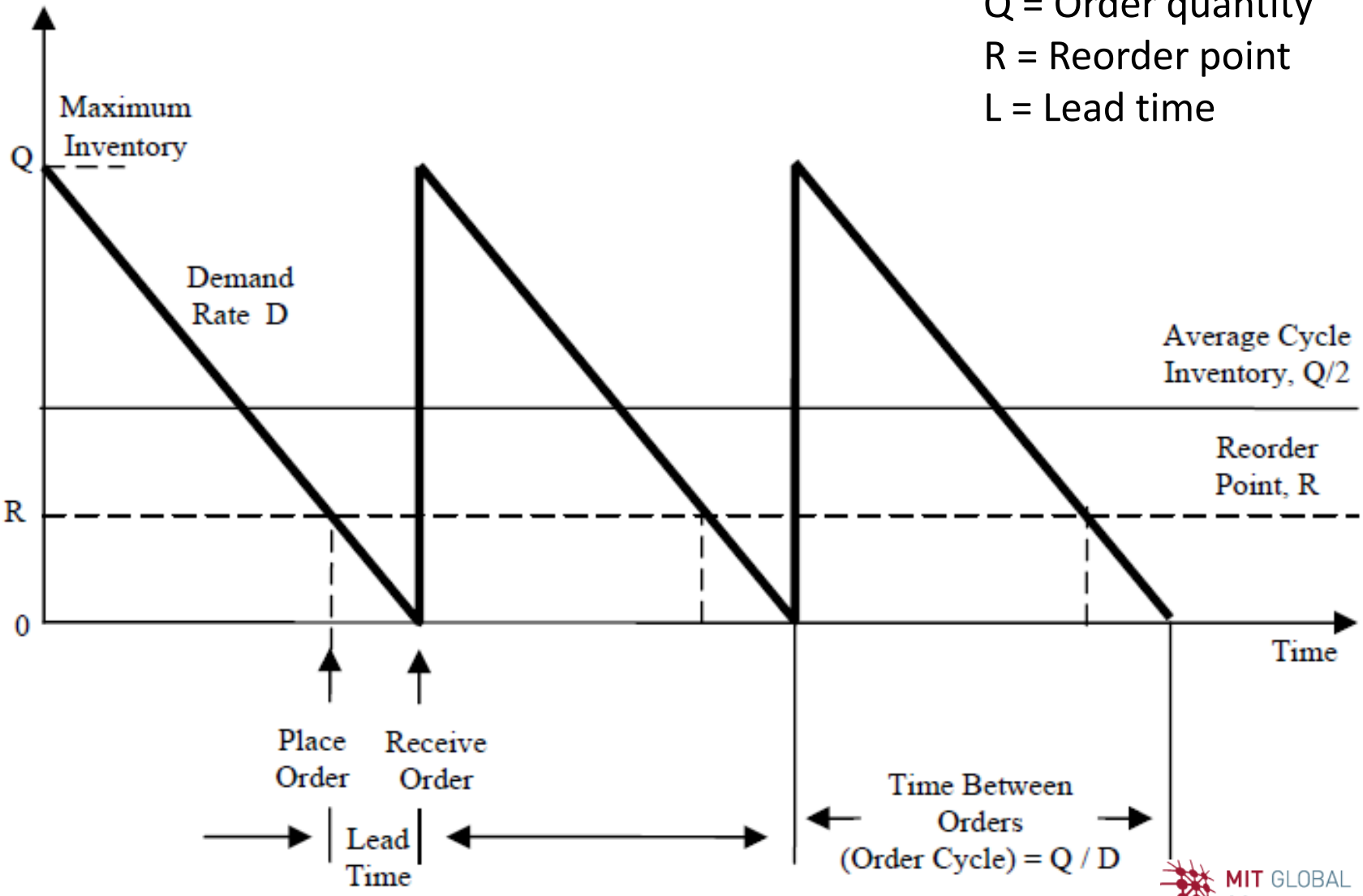
- ✓ Demand for the product is known and constant throughout the planning horizon.
- ✓ Lead time (time from ordering to receipt) is known and constant.
- ✓ Receipt of inventory is instantaneous.
- ✓ Price per unit of product is constant.
- ✓ Inventory holding cost is based on average inventory.
- ✓ Ordering or setup costs per batch are constant (i.e., do not depend on order quantity).
- ✓ All demands for the product will be satisfied (no back orders are allowed).

Pattern of inventory level for R13



EOQ dynamics

D = Demand rate
Q = Order quantity
R = Reorder point
L = Lead time



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Model parameters

Q* = Optimal order quantity (the EOQ)

D = Annual demand

S = Ordering (setup) cost per order

H = Holding (or carrying) cost per unit per year

C = Purchasing cost per unit

Holding cost (H) can be expressed either:

- As a fixed cost, e.g.,

  H = $0.50 per unit per year

- As a percentage (h) of the item's purchase cost (C), e.g.,

  H = h x C = 10% x $5

EOQ model: Total (annual) cost

Total ordering cost = (D/Q) x S
Total holding cost = (Q/2) x H
} Total Variable Costs

Total purchasing cost = C x D

Total annual cost $= C \cdot D + \left(\frac{D}{Q}\right) \cdot S + \left(\frac{Q}{2}\right) \cdot H$

Note:

- (Q/2) is the average inventory level
- Total purchasing cost does not depend on Q

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The Atlantic Coast Tire Case

1. Annual purchasing cost = C x D
2. Annual ordering cost = (D/Q) x S
3. Annual holding cost = (Q/2) x H
4. Annual shortage cost = 0 (no shortage allowed)

$$TAC(Q) = CD + \left(\frac{D}{Q}\right)S + \left(\frac{Q}{2}\right)H$$

$$TVC(Q) = \left(\frac{D}{Q}\right)S + \left(\frac{Q}{2}\right)H$$

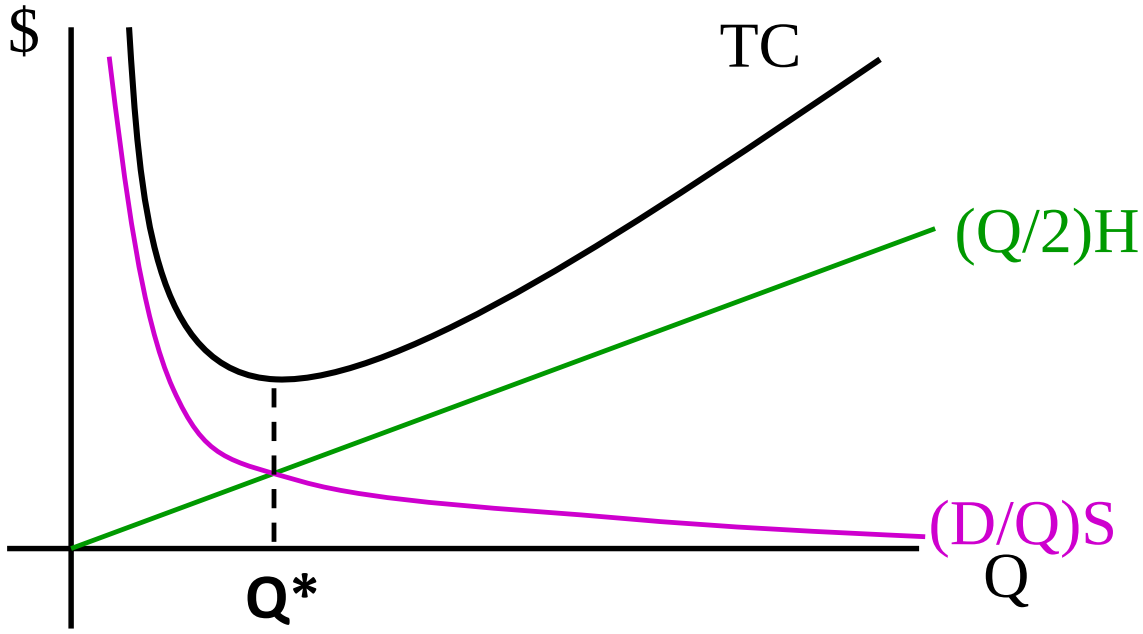
- Under ACT's current policy:
  - TVC = 115*6 + 500*4.20 = $2790
- Is this optimal? What is the Optimal Order Quantity or Economic Order Quantity (EOQ)?

Finding the optimal EOQ

- Only ordering and holding costs need to be minimized (all other costs do not depend on the order quantity)
- As Q (order quantity) increases:
  - Holding cost increases
  - Ordering cost decreases (since the number of orders per year decreases)

Finding the optimal EOQ

Total Variable Cost: $TVC(Q) = \left(\frac{D}{Q}\right)S + \left(\frac{Q}{2}\right)H$



$$Q^* = \sqrt{\frac{2DS}{H}}$$

$$TVC(Q^*) = \sqrt{2DSH}$$

The Atlantic Coast Tire Case

- For ACT's problem,

$$Q^* = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \cdot 6000 \cdot 115}{4.20}} = 573$$

- Which gives TVC

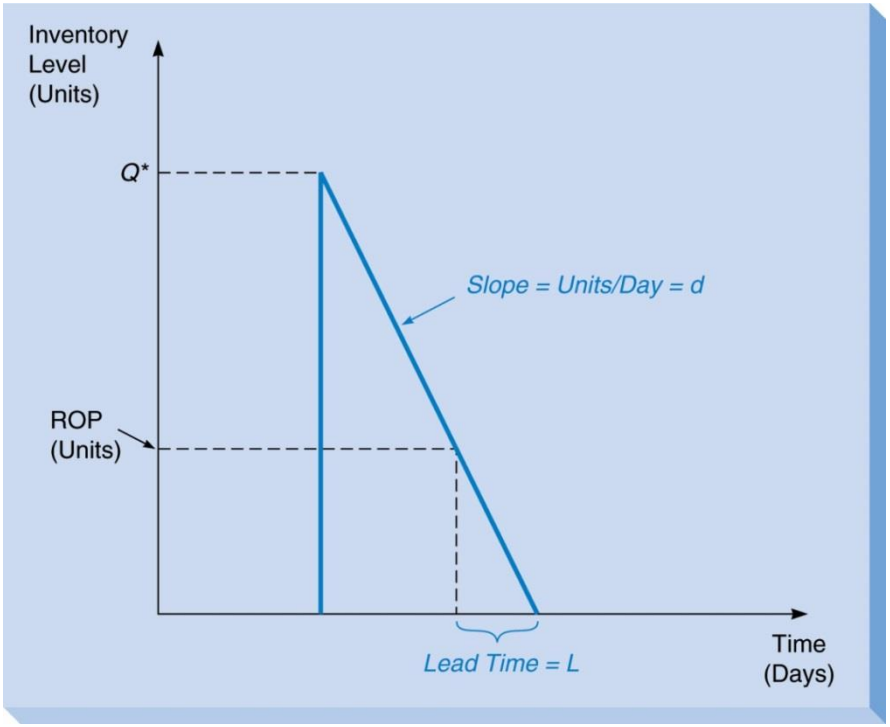
$$TVC(Q^*) = \sqrt{2DSH} = \sqrt{2 \cdot 6000 \cdot 115 \cdot 4.20}$$
$$= \$2407$$

- How does this compare with the current policy?

A 14% reduction

Reorder Point: Determining when to order

- After Q* is determined, the second decision is when to order
- Orders must usually be placed *before* inventory reaches 0 due to order lead time



ROP = (D x L) units

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Example

A manufacturer uses a specific component at a steady rate of 60 per week. The company purchases this component from a supplier at a cost of $0.50 each. The purchasing department estimates that it costs $12.20 to process an order, and the accounting department estimates the cost of carrying inventory at 25% per year.

Example (cont.)

- What are the problem parameters?
  - S = \$12.20
  - C = \$0.50
  - h = 25% per year
  - H = h x C = 0.25 x 0.50 = \$0.125 per year
- Demand rate is given per week but carrying cost is per year => **ALWAYS convert parameter values to a common time unit**
  - D = 60 units/week x 52 weeks/year
    = 3,120 units per year

Example (cont.)

- Optimal order quantity

$$\text{EOQ} = Q^* = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \cdot 3120 \cdot 12.2}{0.125}} = 780 \text{ units}$$

- Cycle time (T)

$$T = \frac{Q^*}{D} = \frac{780}{3120} = 0.25 \text{ years}$$

- Frequency of orders (n)

$$n = \frac{1}{T} = 4 \text{ orders per year}$$

- Annual variable cost

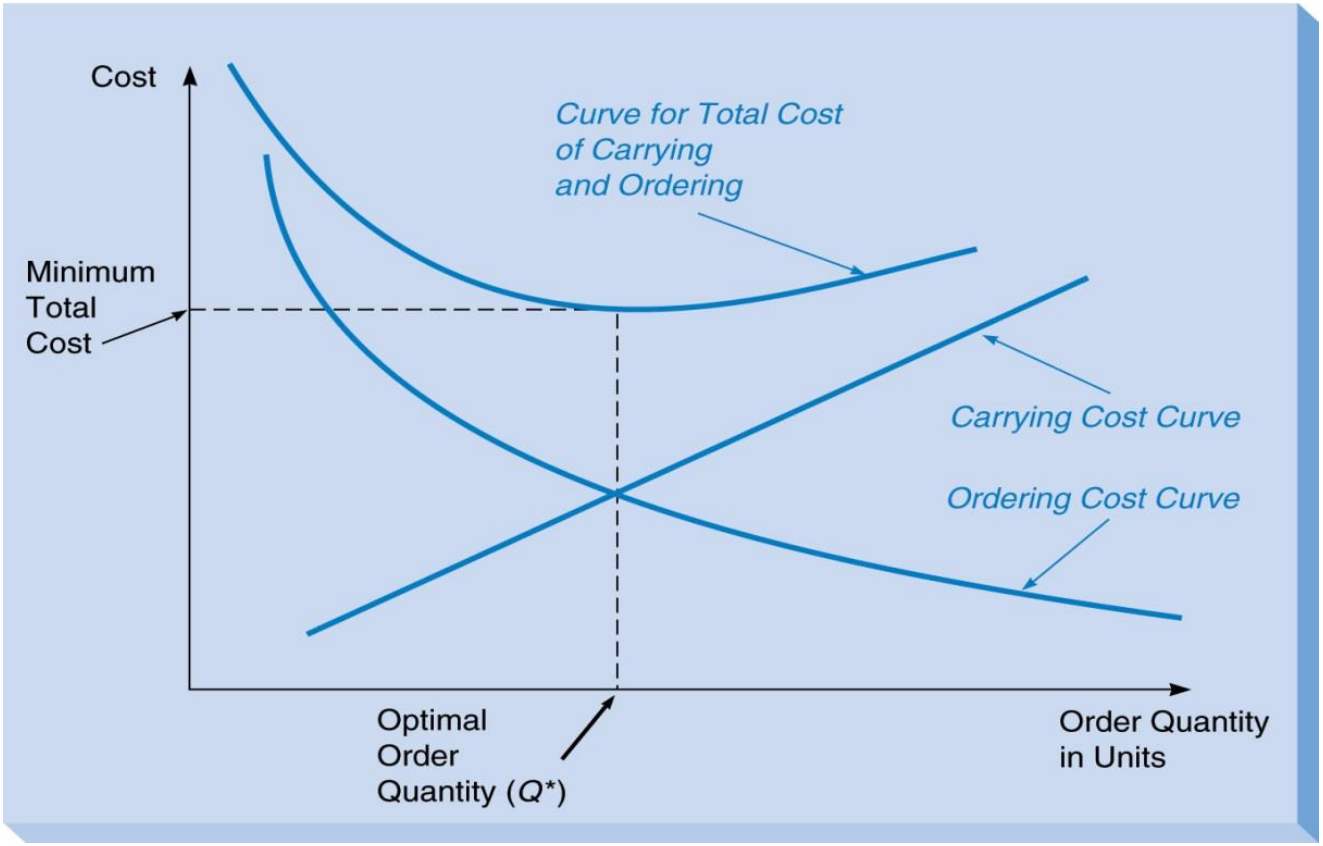
$$TVC(Q^*) = \sqrt{2DSH} = \sqrt{2 \cdot 3120 \cdot 12.2 \cdot 0.125} = \$97.6$$

EOQ sensitivity analysis

- The EOQ model assumes all inputs are known with certainty
- In reality these values are often estimates
- Determining the effect of input value changes on Q* is called **sensitivity analysis**
  - For example, what is the impact on Q* of an increase of holding cost by 50%

$$Q^* = \sqrt{\frac{2DS}{H}}$$

EOQ sensitivity analysis (cont.)



- TVC(Q) is “flat” around EOQ
- Modest deviations from EOQ are not costly

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EOQ sensitivity analysis (cont.)

- EOQ is quite "robust". It can be shown that:

$$\frac{\mathrm{TVC(Q)}}{\mathrm{TVC(Q^*)}} = \frac{1}{2}\left(\frac{\mathrm{Q}}{\mathrm{Q^*}} + \frac{\mathrm{Q^*}}{\mathrm{Q}}\right)$$

- Suppose we overestimate Q* by 1/3; so, Q/Q* = 4/3. Then,

$$\frac{\mathrm{TVC(Q)}}{\mathrm{TVC(Q^*)}} = \frac{25}{24}$$

A **cost penalty** of just over 4%

Basic+EOQ.xlsx

EOQ practical implications

- One difficulty with EOQ is that the optimal reorder interval (T*) may take on any value, which might be impractical from the planning perspective.
- ***Power of two*** policies restrict T to be a power of two multiple of some fixed base planning period T_B; that is, $T = 2^k T_B$.
- Since TVC(T) is convex, the optimal k (i.e., k*) is the smallest integer k satisfying

$$\mathrm{TVC}\left(2^k T_B\right) \leq \mathrm{TVC}\left(2^{k+1} T_B\right)$$

- It can be shown that $\frac{\mathrm{TVC}(2^{k^*}\mathrm{T})}{\mathrm{TVC}(\mathrm{T}^*)} \leq 1.06$

Key points from the EOQ model

- The EOQ is a decision regarding economies of scale.
- EOQ often determines the cycle stock a firm carries.
- In deciding the optimal lot size, the tradeoff is between setup (order) cost and holding cost.
- The total variable cost is relatively “flat” around EOQ; that is, modest deviations around EOQ are not costly.
- If lot size is to be reduced, one has to reduce fixed ordering cost; e.g., to reduce lot size by a factor of 2, ordering cost has to be reduced by a factor of 4.

Thank you!



"Bad news, fellas... it's inventory time."

HW1 will be posted today